

Geographic Aspects of Temperature and Concentration Feedbacks in the Carbon Budget

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ABSTRACT

The geographical distribution of feedback processes in the carbon budget is investigated in a manner that parallels that for climate feedback/sensitivity in the energy budget. Simulations for a range of emission scenarios, made with the Canadian Centre for Climate Modelling and Analysis (CCCma) earth system model (CanESM1), are the basis of the analysis. Anthropogenic CO₂ emissions are concentrated in the Northern Hemisphere and provide the forcing for changes to the atmospheric carbon budget. Transports redistribute the emitted CO₂ globally where local feedback processes act to enhance (positive feedback) or suppress (negative feedback) local CO₂ amounts in response to changes in CO₂ concentration and temperature.

An increased uptake of CO₂ by the land and ocean acts to counteract increased atmospheric CO₂ concentrations so that “carbon–concentration” feedbacks are broadly negative over the twenty-first century. Largest values are found over land and particularly in tropical regions where CO₂ acts to fertilize plant growth. Extratropical land also takes up CO₂ but here the effect is limited by cooler temperatures. Oceans play a lesser negative feedback role with comparatively weak uptake associated with an increase in the atmosphere–ocean CO₂ gradient rather than with oceanic biological activity.

The effect of CO₂-induced temperature increase is, by contrast, to increase atmospheric CO₂ on average and so represents an overall positive “carbon–temperature” feedback. Although the average is positive, local regions of both positive and negative carbon–temperature feedback are seen over land as a consequence of the competition between changes in biological productivity and respiration. Positive carbon–temperature feedback is found over most tropical land while mid–high-latitude land exhibits negative feedback. There are also regions of positive and negative oceanic carbon–temperature feedback in the eastern tropical Pacific.

The geographical patterns of carbon–concentration and carbon–temperature feedbacks are comparatively robust across the range of emission scenarios used, although their magnitudes are somewhat less robust and scale nonlinearly as a consequence of the large CO₂ concentration changes engendered by the scenarios. The feedback patterns deduced nevertheless serve to illustrate the localized carbon feedback processes in the climate system.

1. Introduction

Anthropogenic emissions of CO₂ affect both the energy and carbon balance of the climate system. Boer and Arora (2009, hereafter BA) analyze the response of the global mean carbon budget to anthropogenic emission of CO₂ in terms of “carbon–temperature” and “carbon–concentration” feedbacks. Increasing surface CO₂ concentration in the atmosphere promotes carbon uptake by the underlying land and ocean and this acts to counteract the atmospheric CO₂ increase and so acts as

a negative feedback in this sense. By contrast, increasing surface temperature promotes the release of CO₂ into the atmosphere and so is a positive feedback. The analysis of carbon feedbacks parallels that for the global “climate feedback/sensitivity,” which relates the global mean temperature response to the radiative forcing due to greenhouse gases (and other forcing agents) in the energy budget. The analyses in the two cases are based on a linearization of the respective carbon and energy budgets.

In the more familiar climate feedback case, Boer and Yu (2003a, hereafter BY) make the argument that the pattern of temperature response does not necessarily resemble that of the radiative forcing itself. The climate system distributes the forcing globally and local feedback processes act to enhance or suppress local temperature

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change. Local radiative forcing may thus give rise to remote temperature responses in regions where strong feedbacks operate, such as in the Arctic in conjunction with the ice–albedo feedback. The geographic pattern of these feedback processes in the energy budget may be diagnosed in terms of the vertically integrated energy budget equation as done in BY, and subsequently in Boer and Yu (2003b), Boer et al. (2005), Stowasser et al. (2006), Webb et al. (2006), and elsewhere.

The pattern of CO₂ concentration change in the atmosphere is a consequence of localized forcing in the form of anthropogenic CO₂ emissions, of atmospheric transports that redistribute this CO₂ globally, and of local feedback processes on the land and in the ocean that act to enhance or decrease the atmospheric CO₂ concentration. For carbon, the temperature change that results from an increase in atmospheric CO₂ also affects the carbon budget through temperature-dependent biological and physical processes.

As discussed in BA, the linearization of the globally averaged atmospheric carbon budget in terms of global mean temperature and concentration change is not as successful as in the case of the energy budget. In particular, while the positive global carbon–temperature feedback is reasonably linear and consistent as temperature increases, the carbon–concentration feedback is not and weakens with increasing CO₂ concentration, thereby enhancing atmospheric CO₂ and accelerating global warming. Moreover, the behavior of the carbon–concentration feedback is different for different emission scenarios so there is some dependence on state variables other than CO₂ concentration. While carbon budget feedbacks, quantified in this way, are not as robust as in the case of energy budget feedbacks, they are nevertheless useful for understanding and quantifying aspects of carbon budget behavior.

Here we extend the analysis of globally averaged carbon budget feedbacks to an analysis of their spatial patterns following the approach of BY for the energy budget. As in the case of the energy budget, local forcings give rise to remote responses. The carbon budget forcing due to anthropogenic CO₂ emissions is found mainly over localized regions of Northern Hemisphere land, the emitted CO₂ is transported throughout the system, and carbon budget responses are enhanced or damped by local feedback processes.

2. Energy and carbon feedbacks

Following BY, quantities are expressed in the form $X = X_o + X'$, where X_o represents the unperturbed climate value and X' is the change due to the external forcing. The vertically integrated energy budget equation is written as

$$\frac{dh'}{dt} = A' + R' = A' + \Lambda T' + f, \quad (1)$$

where dh'/dt is the change in the heat content of the system, A' is the change in the transport term representing the convergence of energy into the column, and R' is the change in the radiative flux into the system. Here R' is further decomposed into a radiative forcing term $f = f(C', \dots)$ due to the increased concentration of CO₂ (C'), other greenhouse gases, aerosol loadings, etc., and a (signed) feedback term $\Lambda T'$, which is expressed as a linear function of surface temperature change. In (1) all quantities $X(\lambda, \varphi)$ are functions of geographical location including the feedback parameter $\Lambda(\lambda, \varphi)$, which must be negative on average if the system is to be stable. For a new equilibrium $dh'/dt = 0$ and $T' = (A' + f)/(-\Lambda) = s(A' + f)$, formally at least, indicating how the resulting temperature pattern is a consequence of the forcing and of its advective redistribution that is localized by the sensitivity parameter s , which is inversely proportional to the feedback parameter. Taking the global average of (1), represented by angular brackets, gives $d\langle h' \rangle / dt = \langle R' \rangle = \hat{\Lambda} \langle T' \rangle + \langle f \rangle$, which becomes $\langle T' \rangle = \langle f \rangle / (-\hat{\Lambda}) = \hat{s} \langle f \rangle$ at equilibrium with the inverse of the global feedback parameter $\hat{\Lambda} = \langle \Lambda T' \rangle / \langle T' \rangle$, providing the linear proportionality between forcing and temperature response.

In BY, information on the geographic distribution of the feedbacks, and indirectly on the physical processes involved, are obtained from the “local contribution” to the global feedback given by

$$\Lambda_l = \frac{\Lambda T'}{\langle T' \rangle} = \frac{(R' - f)}{\langle T' \rangle}, \quad (2)$$

where the global average of the local feedback contribution is the global feedback parameter $\hat{\Lambda} = \langle \Lambda_l \rangle$. The feedback parameter Λ_l may be further decomposed into components associated with different processes that have a signature in the radiative flux change R' .

The vertically integrated atmospheric carbon budget equation may be expressed in a form analogous to (1) as

$$\frac{dH'}{dt} \approx m \frac{dC'}{dt} = \mathcal{A}' + F' + E = \mathcal{A}' + \Gamma T' + BC' + E. \quad (3)$$

Here $H' \approx mC'$ is the change in the amount of carbon in the atmospheric column, which is approximately the change in surface CO₂ concentration C' times the mass m of the atmosphere in the column since CO₂ is reasonably well mixed. Again $\mathcal{A}'$ is the change in the transport term in this case of CO₂, F' is the change in the flux of CO₂ into the atmosphere from the surface

beneath, and E is the anthropogenic emissions (which are zero for the unperturbed climate). The response $F' = \Gamma T' + BC'$ to the forcing E is expressed as a linear function of changes in temperature and CO_2 concentration paralleling (1) where terms are again functions of geographical position. Equations (1) and (3) are linked equations for temperature and CO_2 concentration. For the global average

$$\frac{d\langle H' \rangle}{dt} = \langle F' \rangle + \langle E \rangle = \hat{\Gamma} \langle T' \rangle + \hat{B} \langle C' \rangle + \langle E \rangle, \quad (4)$$

and, as in (2), Γ_l and B_l are the local contributions to the global feedback parameters with

$$\Gamma_l = \frac{\Gamma T'}{\langle T' \rangle}, \quad B_l = \frac{BC'}{\langle C' \rangle} \quad (5)$$

and where $\hat{\Gamma} = \langle \Gamma_l \rangle$, $\hat{B} = \langle B_l \rangle$.

The two feedback parameters Γ and B involved in (4) and (5) are evaluated from the results of three simulations as described in BA and following the Coupled Carbon Cycle Model Intercomparison Project (C^4MIP) approach (e.g., Friedlingstein et al. 2006). The three simulations include the usual *control* simulation with no anthropogenic emissions of CO_2 , a *climate change* simulation with specified CO_2 emissions, and a *radiatively uncoupled* simulation where the radiation code in the model sees only the control run value of CO_2 even though emissions are nonzero and the carbon cycle is operating. The budget paralleling (3) for the radiatively uncoupled case is

$$\begin{aligned} \frac{dH^*}{dt} &\approx m \frac{dC^*}{dt} = \mathcal{A}^* + F^* + E \\ &\approx \mathcal{A}^* + \Gamma T^* + BC^* + E, \end{aligned} \quad (6)$$

where $X^* = X_u - X_o$ represents a difference from the control simulation. Because there is no CO_2 -induced warming in this case the temperature change T^* , although not exactly zero, is small. The flux differences from the control simulation for the radiatively uncoupled and fully coupled cases are approximated as

$$\begin{aligned} F^* &= F_u - F_o \approx \Gamma T^* + BC^* \\ F' &= F - F_o \approx \Gamma T' + BC'. \end{aligned} \quad (7)$$

In BA, the globally averaged version of (7), written here as

$$\begin{aligned} \langle F^* \rangle &\approx \hat{\Gamma} \langle T^* \rangle + \hat{B} \langle C^* \rangle \\ \langle F' \rangle &\approx \hat{\Gamma} \langle T' \rangle + \hat{B} \langle C' \rangle, \end{aligned} \quad (8)$$

is solved for the global feedback parameters $\hat{\Gamma}$, $\hat{B}$. Substituting the local contribution terms (5) into (7) and solving gives the corresponding local feedback contributions as

$$\begin{aligned} \Gamma_l &= \frac{\langle C^* \rangle F' - \langle C' \rangle F^*}{\langle T' \rangle \langle C^* \rangle - \langle T^* \rangle \langle C' \rangle} \\ B_l &= \frac{\langle T' \rangle F^* - \langle T^* \rangle F'}{\langle T' \rangle \langle C^* \rangle - \langle T^* \rangle \langle C' \rangle}, \end{aligned} \quad (9)$$

which satisfies (8) under global averaging and for which $\hat{\Gamma} = \langle \Gamma_l \rangle$, $\hat{B} = \langle B_l \rangle$.

3. CanESM1 and carbon cycle simulations

The first version of the Canadian Centre for Climate Modeling and Analysis (CCCma) Earth System Model (CanESM1) is a carbon climate model based on the third-generation CCCma coupled atmosphere–ocean climate model (CGCM3), which provided output for the recent Intergovernmental Panel on Climate Change (IPCC) assessment report (Solomon et al. 2007). Terrestrial ecosystem processes are modeled using the Canadian Terrestrial Ecosystem Model (CTEM; Arora 2003; Arora and Boer 2003, 2005), which uses a single-leaf photosynthesis model based on the biochemical model of Collatz et al. (1991, 1992). The coupling between photosynthesis and canopy conductance is based on vapor pressure deficit (Leuning 1995). Autotrophic respiratory fluxes from vegetation are functions of temperature, while heterotrophic respiratory fluxes from dead carbon pools of litter and soil carbon are functions of both moisture and temperature. The oceanic ecosystem and inorganic carbon processes are modeled using the Canadian Model of Ocean Carbon (CMOC; Zahariev et al. 2008). The inorganic chemistry module is based on protocols from the Ocean Carbon Model Intercomparison Project (OCMIP) Phase 2 (<http://www.ipsl.jussieu.fr/OCMIP/phase2/>), with dissolved inorganic carbon (DIC) and total alkalinity as prognostic variables. The ecosystem component of CMOC is based on the nitrogen–phytoplankton–zooplankton–detritus (NPZD) model of Denman and Peña (1999). Detailed results from CanESM1 for the simulation of the carbon budget of the preindustrial control climate are given by Christian et al. (2009, manuscript submitted to *J. Geophys. Res.*) and for the twentieth-century carbon budget by Arora et al. (2009).

We consider three emission scenarios; the A2 simulation uses CO_2 emissions based on the Special Report on Emissions Scenarios (SRES) A2 scenario, the corresponding B1 simulation uses emissions based on the SRES B1 scenario, and the E6 simulation has constant CO_2 emissions of 6 PgC yr^{-1} . Other greenhouse gases (GHGs) and aerosols remain at their control run values. Radiatively uncoupled and fully coupled simulations are performed in each case for the period from 1850 to 2000

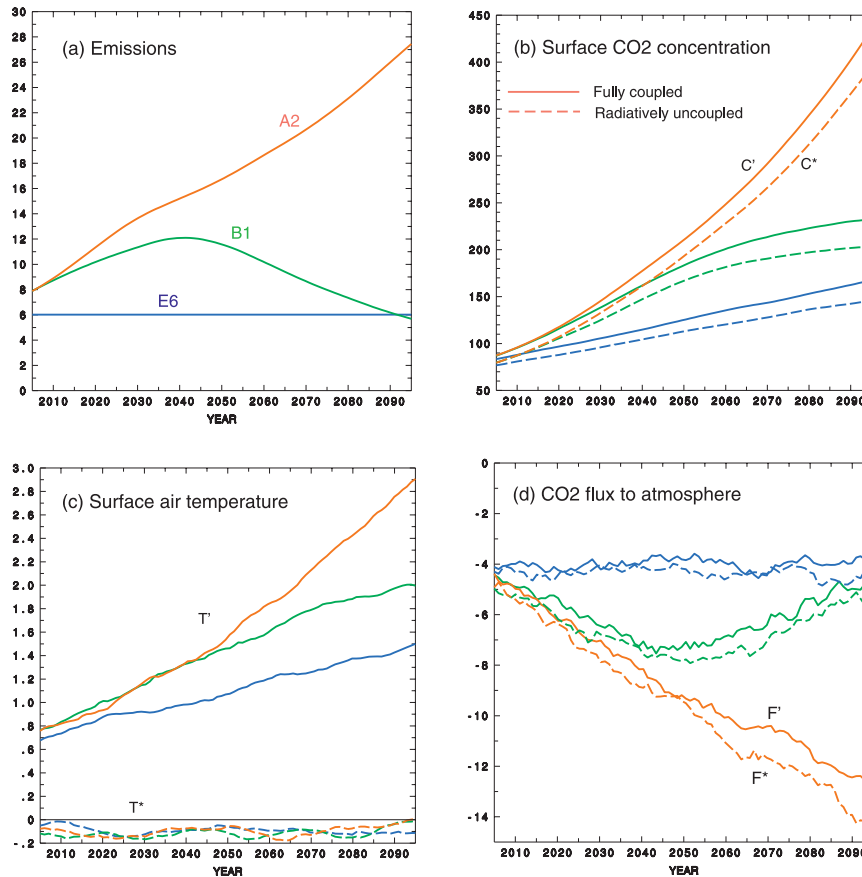


FIG. 1. (a) Emission rates (PgC yr^{-1}) for the E6 constant emission scenario and for the B1 and A2 SRES scenarios, (b) surface CO_2 concentration differences (ppmv) from preindustrial values for the radiatively uncoupled C^* and fully coupled C' cases, (c) surface air temperature differences T^* and T' ($^{\circ}\text{C}$), and (d) atmospheric CO_2 flux differences F^* and F' (PgC yr^{-1}). Blue, green, and orange results are from simulations using the E6, B1, and A2 emission scenarios. Results have been smoothed with an 11-yr running mean filter.

using historical emissions followed by scenario emissions for the period 2001–2100. The analysis concentrates on the means of the simulations at the end of this period, namely from 2070 to 2100.

4. Temperature and CO_2 changes and feedbacks

a. Global values

Figure 1 gives globally averaged changes in quantities entering the carbon budget over the twenty-first century as presented in BA. Emissions for the A2 scenario (orange lines) increase more or less linearly over the period while the B1 emissions (green) peak more or less in the middle of the period and, of course, the E6 emissions (blue) are constant. The corresponding CO_2 concentrations increase more or less linearly for the E6 simulation and are convex upward for the A2 and concave for the B1 simulations. Temperatures increase at different rates depending on scenario as do CO_2 fluxes out of the

atmosphere. Results for the fully coupled simulations are the solid lines in the figure and those for the radiatively uncoupled simulations are dashed.

In Fig. 1d, $F^* = F_u - F_o$ is the change in CO_2 flux into the atmosphere due to an increase in atmospheric CO_2 but in the absence of a corresponding CO_2 -induced temperature change (T^* is small in the radiatively uncoupled simulations in Fig. 1c). The system responds to the increasing atmospheric CO_2 concentration by enhanced uptake of CO_2 by the land and ocean, hence increasing the rate of loss from the atmosphere. Since these processes counteract the increase in atmospheric CO_2 concentration, they constitute a negative “carbon-concentration” feedback. Here F' is the flux change in the presence also of the CO_2 -induced temperature change T' and $F' - F^*$ gives the response of the CO_2 budget to this temperature increase. In Fig. 1d, $F' - F^* > 0$ indicating a positive “carbon-temperature” feedback acting to promote CO_2 flux into the atmosphere

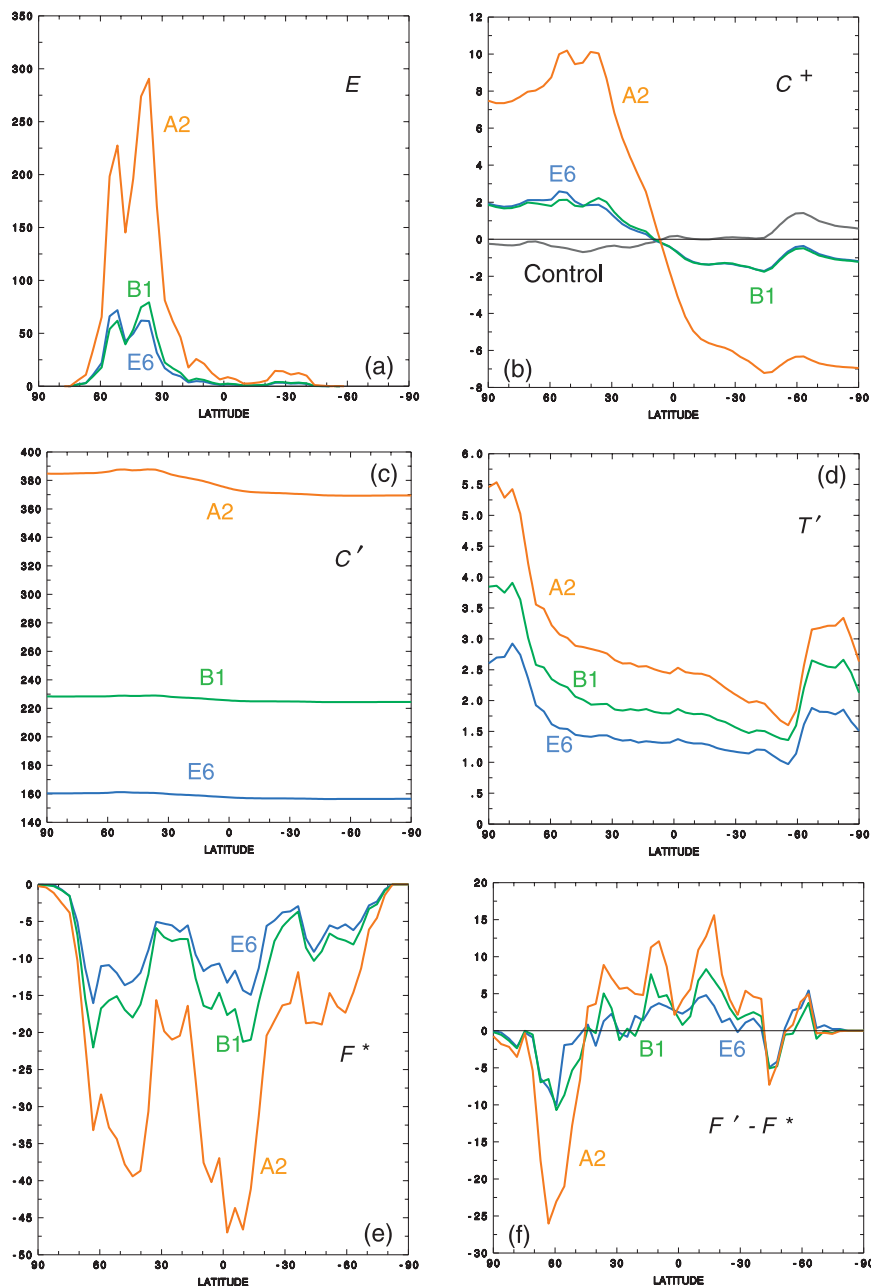


FIG. 2. Zonal averages of quantities averaged over years 2070–2100 of the simulations forced with the E6, B1, and A2 emission scenarios. (a) Emission rates E ($\text{gC m}^{-2} \text{yr}^{-1}$), (b) CO_2 concentrations C^+ as deviations from the global mean (ppmv), (c) CO_2 concentration changes C' for fully coupled simulations (ppmv), (d) associated temperature changes T' ($^{\circ}\text{C}$), (e) the change in fluxes into the atmosphere F^* for the radiatively uncoupled simulation ($\text{gC m}^{-2} \text{yr}^{-1}$), and (f) the difference $F' - F^*$ with the fully coupled simulation ($\text{gC m}^{-2} \text{yr}^{-1}$). Blue, green, and orange results apply to results from simulations using the E6, B1, and A2 emission scenarios.

(by reducing the flux out of the atmosphere in this case) as temperatures warm. Increasing temperature decreases, on average, the ability of the land and ocean to absorb emitted CO_2 .

b. North–south structures

Figure 2 displays the meridional structures of terms in the atmospheric carbon budget averaged over years

TABLE 1. Spatial standard deviations and correlations of year 2070–2100 averaged fields.

Scenario	Spatial standard deviations			Spatial correlations		
	A2	B1	E6	$r(A2, B1)$	$r(A2, E6)$	$r(B1, E6)$
T' (°C)	0.96	0.69	0.52	0.97	0.95	0.97
C^* (ppmv)	7.90	2.12	2.07	0.99	0.99	0.99
C' (ppmv)	7.52	1.89	1.87	0.99	0.98	0.98
F^* (gC m ⁻² yr ⁻¹)	41.9	20.3	15.0	0.91	0.84	0.86
F' (gC m ⁻² yr ⁻¹)	39.5	19.4	14.6	0.90	0.82	0.83
B_l (gC m ⁻² yr ⁻¹ ppmv ⁻¹)	0.12	0.10	0.10	0.91	0.85	0.87
Γ_l (gC m ⁻² yr ⁻¹ °C ⁻¹)	9.35	7.58	8.29	0.67	0.61	0.46

2070–2100 of the simulations. By the end of the simulation period, the E6 and B1 emission rates in Fig. 2a are similar while the A2 emission rate is considerably stronger. Not surprisingly, the change in CO₂ concentration C' in Fig. 2c is largest for the A2 scenario followed by that for the B1 and E6 scenarios. Even though the emission rates for B1 and E6 are similar at the end of the simulation period, the B1 concentration reflects the larger amount of CO₂ emitted into the atmosphere over the period to 2100. For the control simulation, the CO₂ concentration displays a weak interhemispheric CO₂ gradient in Fig. 2b with somewhat larger values in the Southern compared to the Northern Hemisphere as highlighted by removing the global mean and plotting $C^+ = C - \langle C \rangle$. For the scenario simulations, with emissions concentrated in the Northern Hemisphere, the interhemispheric gradient reverses and strengthens, with the gradient for the A2 scenario appreciably larger than that for the B1 and E6 scenarios. The latitudinal distribution of the temperature change T' in Fig. 2d displays the expected result, with larger warming at polar latitudes and in the Northern versus the Southern Hemispheres.

Figure 2 also displays the zonal averaged values of the CO₂ flux between the atmosphere and the underlying surface. In Fig. 2e, $F^* = F_u - F_o$ indicates how the carbon budget reacts to an increase of CO₂ concentration in the absence of a temperature change. As the CO₂ concentration increases because of the emissions, the atmosphere loses CO₂ to the underlying surface at a rate

that depends on location and concentration change (and possibly other state variables). The effect is to counteract the increase in CO₂ in the atmosphere and so the carbon–concentration feedback is a negative in this sense. The sign of the temperature feedback on the atmospheric carbon budget is indicated by $F' - F^*$ in Fig. 2f and, although the global average carbon–temperature feedback is positive, $F' - F^*$ displays both positive values at lower latitudes and negative values at higher latitudes, notably in the Northern Hemisphere.

c. Geographical distributions

The geographical distributions of the changes to carbon budget terms averaged over years 2070–2100 of the simulations are plotted for the strongly forced A2 simulation. They are compared with the results from the other simulations in terms of second-order statistics in Table 1. The spatial standard deviation of the time averaged field X is $\sigma_X = \sqrt{\langle X^{+2} \rangle}$, for $X^+ = X - \langle X \rangle$, which is the deviation from the global average. The spatial standard deviation serves to quantify the spatial “amplitudes” of the geographic patterns, while the agreement between the spatial patterns of two fields is measured by the spatial correlation $r_{X,Y} = \langle X^+ Y^+ \rangle / \sigma_X \sigma_Y$. Table 1 compares results for a given variable across the three scenarios, while Table 2 compares results across variables for the A2 scenario.

Figure 3 (upper panel) gives the preindustrial distribution of surface CO₂ concentration C_o , which has relatively

TABLE 2. Spatial correlations between year 2070–2100 averaged fields for the A2 scenario.

	Γ_l	T'	C^*	C'	$C' - C^*$	F^*	F'	$F' - F^*$	E
B_l	−0.51	−0.23	−0.08	−0.11	−0.19	1.00	0.85	−0.37	−0.24
Γ_l		0.06	−0.11	−0.07	0.55	−0.52	0.03	0.99	0.07
T'			0.40	0.39	−0.31	−0.23	−0.23	0.03	0.14
C^*				1.00	−0.64	−0.08	−0.16	−0.13	0.50
C'					−0.59	−0.10	−0.17	−0.10	0.52
$C' - C^*$						−0.19	0.12	0.56	−0.10
F^*							0.84	−0.38	−0.24
F'								0.18	−0.23
$F' - F^*$									0.03

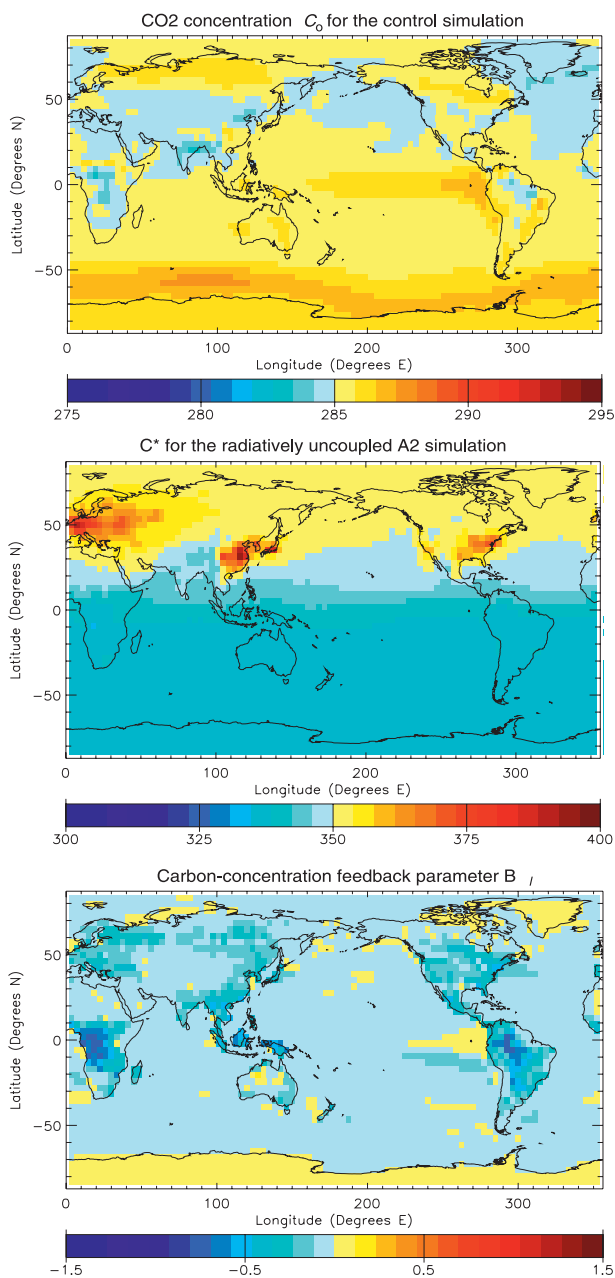


FIG. 3. (top) CO₂ concentration C_0 (ppmv) for the preindustrial control climate; (middle) the change in CO₂ concentration C^* (ppmv) for years 2070–2100 of the radiatively uncoupled A2 emission scenario simulation; (bottom) the carbon–concentration feedback parameter B_1 (gC m⁻² yr⁻¹ ppmv⁻¹).

little spatial structure. The global mean is 285.4 ppmv so the warm (cool) colors broadly indicate regions with concentrations above (below) the global mean. Concentrations are slightly higher in the Southern Hemisphere. Patches of comparatively low CO₂ are seen over low-midlatitude land and midlatitude oceans.

The middle panel of Fig. 3 displays the change in surface CO₂ concentration C^* for the radiatively uncoupled A2 simulation where the localized emission sources, identified by localized maxima in C^* , have reversed the interhemispheric CO₂ gradient (shown as C^+ in Fig. 2b). In this radiatively uncoupled case the temperature change T^* is small and the CO₂ distribution may nominally be written from (6) as $C^* \approx (A^* + E - dH^*/dt)/(-B) = S(A^* + E - dH^*/dt)$, illustrating the balance between the local emissions, the transport away from these localized sources, and the rate of change of storage in the atmosphere, all scaled by the local carbon–concentration sensitivity parameter S , which is the inverse of the negative feedback parameter B . In regions remote from the emission sources, $E = 0$ and the concentration depends mainly on the transport and the local feedback/sensitivity. The resulting distribution of C^* clearly reflects this balance with largest concentrations in the regions of the sources but with the CO₂ distributed throughout the system by the transport. The spatial standard deviations of C^* in Table 1 are largest for the more strongly forced A2 simulation and weakest for the least strongly forced E6 simulation, as might be expected. Although emissions, global CO₂ concentrations, and spatial standard deviations differ among the scenarios the resulting spatial patterns of CO₂ change are very similar with spatial correlation coefficients that equal or exceed 0.98 in Table 1.

The contribution of local feedbacks to the global carbon–concentration feedback is quantified by the feedback parameter B_1 in the lower panel of Fig. 3 and, according to (9), will essentially be a scaled version of F^* indicating how the surface fluxes respond to an increase in CO₂ concentration in the absence of temperature change. Largest feedbacks are seen over land and especially tropical land where biological productivity is high (e.g., Hickler et al. 2008). The feedback parameter is a measure of the additional biological productivity, acting as a negative feedback to atmospheric CO₂, which results as a consequence of the fertilization effect of increasing CO₂ concentration. Clearly these feedbacks depend on local processes and are, in most cases, far removed from the emission sources. In fact, the common spatial variance of B_1 and E is only about 6% [$r^2(B_1, E) = (-0.24)^2 = 0.06$ from Table 2]. Carbon–concentration feedback is reasonably strong over much of midlatitude North America and Asia, while lack of moisture forestalls CO₂ uptake over Australia, Saharan Africa, and the dry regions of Asia. At high latitudes where temperatures are cold for much of the year, the increased CO₂ fertilization effect is less effective. Although terrestrial carbon–concentration feedbacks are stronger, there is also some evidence of both negative and positive carbon–concentration feedbacks in the tropical east Pacific. Finally, the pattern of

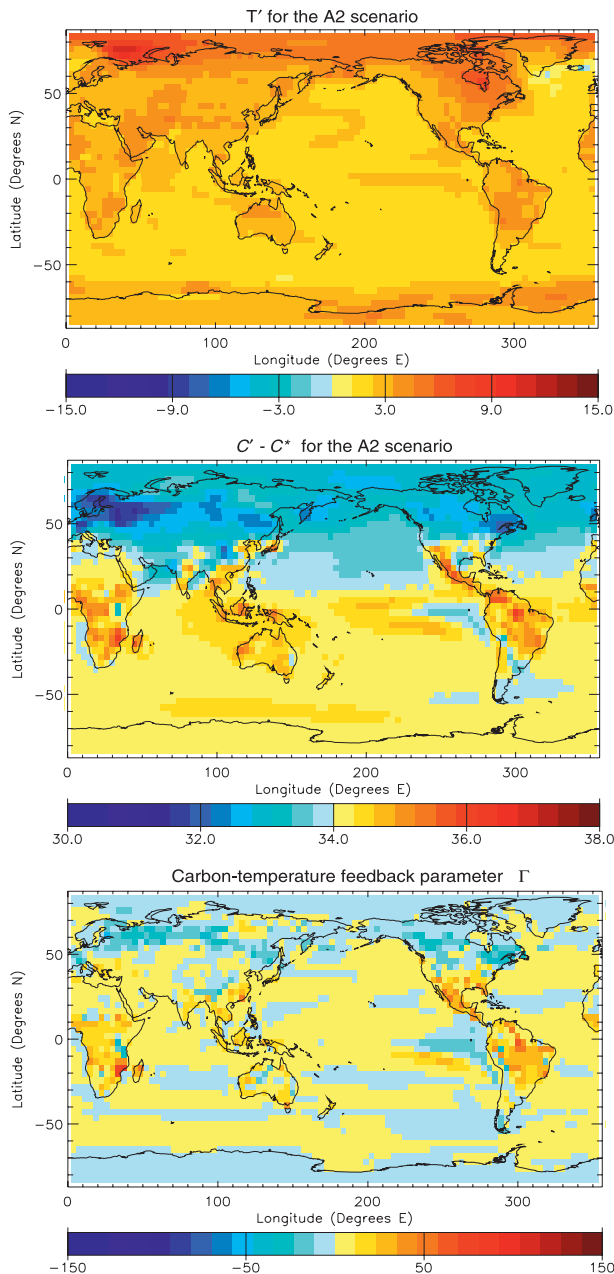


FIG. 4. (top) The temperature change T' ($^{\circ}\text{C}$) for years 2070–2100 for the fully coupled A2 emission scenario simulation; (middle) the difference $C' - C^*$ in the CO_2 concentration as a consequence of this temperature change as it affects the carbon budget; (bottom) the carbon–temperature feedback parameter Γ_l ($\text{gC m}^{-2} \text{yr}^{-1} \text{ } ^{\circ}\text{C}^{-1}$).

the feedback parameter B_l is similar for the different scenarios according to the spatial correlation coefficients in Table 1, which range from 0.85–0.91.

Figure 4 gives the temperature change T' for the A2 simulation averaged over years 2070–2100. Results are more or less standard for global warming calculations,

with the largest warmings over Northern Hemisphere land and where ice and snow have retreated. The spatial standard deviations of T' in Table 1 are largest for the more strongly forced A2 simulation and weakest for the least strongly forced E6 simulation, while the patterns of the temperature change are very similar as measured by the spatial correlations.

The effect of this temperature change on the CO_2 distribution is shown by the difference $C' - C^*$ in the middle panel of Fig. 4. Globally this is reflected in the difference between the orange CO_2 concentration curves in Fig. 1b over the last 30 years of the period, while Fig. 4 gives the geographic pattern of this difference. The overall effect of the temperature increase is to increase the concentration of CO_2 in the atmosphere and it is a positive feedback in this sense. This increase is not spatially uniform and increases are generally retarded in mid–high latitudes of the Northern Hemisphere compared to tropical and Southern Hemispheric areas as indicated by the cool and warm colors in the diagram. Temperature increases can stimulate plant growth, with the resulting CO_2 uptake a negative feedback on atmospheric CO_2 but warmer temperatures also enhance respiration, which returns CO_2 to the atmosphere and is a positive feedback mechanism. Despite the fact that $C' - C^*$ is the consequence of the temperature change T' , the two fields are not strongly spatially correlated in Table 2 and, in fact, are anticorrelated. This indicates once again the local nature of the carbon–temperature feedbacks involved.

The carbon–temperature feedback parameter Γ_l , plotted in the lower panel of Fig. 4, largely reflects the flux difference $F' - F^*$ according to (9) as a measure of the effect of temperature on these fluxes. Here Γ_l displays regions of both positive and negative values over land and ocean. In particular, tropical land is a region of generally positive feedback where increasing temperatures provide CO_2 to the atmosphere in response to modest warming. Mid- and high-latitude land are areas of negative feedback where warming enhances CO_2 uptake and reduces atmospheric CO_2 . Feedback is generally weak over the oceans although it is positive on average. Somewhat stronger regions of positive and negative carbon–temperature feedback Γ_l are seen in the tropical eastern Pacific and are of opposite in sign to the carbon–concentration feedback B_l there. The feedback parameters B_l and Γ_l are spatially anticorrelated according to Table 2. Thus, while carbon–concentration feedbacks are almost everywhere negative, and when positive are small in magnitude, carbon–temperature feedbacks have important contributions of both signs in different geographical regions and especially over land.

Yoshikawa et al. (2008) analyze the carbon–temperature effect directly in terms of carbon fluxes rather than via

the feedback formalism of (6)–(9) but do not consider carbon–concentration feedback. Their Fig. 3a for the land and their Fig. 7 for the ocean should resemble the pattern of Γ_l in Fig. 4 (although with reversed signs). Visually, at least, the correspondence is only fair, implying that the carbon feedbacks in the two models may differ substantially and suggesting the importance of a better understanding of geographical aspects of both carbon–temperature and carbon–concentration feedbacks.

The spatial patterns of B_l are comparatively robust across the results of the three simulations as seen from Table 1 where spatial correlations range from 0.91 to 0.87. The spatial pattern of the carbon–temperature feedback Γ_l is weaker in structure than that of the carbon–concentration feedback, as measured by their spatial standard deviations, and the spatial correlations display less robust agreement across simulations with values ranging from 0.67 to 0.46. As discussed in BA, the global carbon–temperature feedback parameter $\bar{\Gamma}$ is comparatively constant and independent of system state, while the behavior of the global feedback parameter $\bar{B}$ indicates that carbon–concentration feedback is nonlinear and depends on system history and state. Since the geographically distributed feedback contributions Γ_l and B_l have been constructed so as to average to these global values they must share in this general behavior.

The coupling of the energy and carbon budgets is particularly apparent if they are expressed in terms of temperature and concentration changes and associated feedback parameters as is done in (1) and (3). If the feedback parameters were reasonably constant and if the system were to come into equilibrium or quasi equilibrium so that the rate of change terms were small, it would nominally be possible to solve explicitly for C' and T' . CO_2 concentration changes are comparatively large, however, and this mitigates against the first-order expansion of the carbon–concentration feedback or the logarithmic CO_2 radiative forcing term in the energy equation. For this reason the feedback parameters in Figs. 3 and 4 depend, to some extent, on the particular emission scenario involved. Their geographical distribution and relative magnitudes nevertheless give considerable insight into the nature of the carbon feedbacks operating in the climate system.

5. Summary

The global carbon feedback analysis of BA is extended to geographical aspects of carbon–concentration and carbon–temperature feedbacks following an approach previously applied to geographical aspects of climate feedback sensitivity in BY. Simulation results obtained with CanESM1, the first version of the CCCma

earth system model (Christian et al. 2009, manuscript submitted to *J. Geophys. Res.*; Arora et al. 2009), are analyzed for three CO_2 emission scenarios, namely for a constant emission scenario of 6 PgC yr^{-1} (the E6 scenario) and for the SRES B1 and A2 emission scenarios. The changes in CO_2 concentrations, temperatures and surface–atmosphere CO_2 fluxes and the associated carbon–concentration and carbon–temperature feedback parameters are obtained, displayed, and compared.

The anthropogenic CO_2 emissions that provide the forcing for the atmospheric carbon budget are largely concentrated in regions of the Northern Hemisphere. Atmospheric transports redistribute the emitted CO_2 globally where local feedback processes act to enhance (positive feedback) or suppress (negative feedback) local CO_2 concentrations. Feedback processes are investigated following the C⁴MIP approach (Friedlingstein et al. 2006), which involves three simulations: namely, a control simulation with no anthropogenic CO_2 emissions, a radiatively uncoupled simulation where CO_2 increases do not affect the carbon budget via temperature changes, and a fully coupled simulation where all feedbacks are operating.

BA show that for the time evolution of the global average carbon budget there is a positive carbon–temperature feedback operating that is reasonably stable, similar, and linear in global temperature change for the three scenarios. The corresponding negative carbon–concentration feedback parameter, however, differs for the three scenarios and is nonlinear in global concentration change. The feedback results nevertheless characterize the broad nature of carbon budget feedbacks and provide insight into the processes involved.

Although the global distribution of CO_2 in the control simulation is spatially rather uniform, there are regions of enhanced concentration over the Southern Ocean and in the tropical Pacific. Anthropogenic CO_2 emissions concentrated over western Europe, the eastern United States, and regions in Asia in the scenarios inject large amounts of CO_2 into the atmosphere, which reverses the interhemispheric CO_2 gradient. The system responds to an increase in atmospheric CO_2 by enhanced surface uptake, which counteracts the atmospheric CO_2 increase, hence exhibiting a negative carbon–concentration feedback. These negative carbon–concentration feedbacks are largest over land and particularly in tropical regions where CO_2 acts to fertilize plant growth. Extratropical land also takes up CO_2 but the fertilization effect is limited by cooler temperatures. The oceans generally display negative carbon–concentration feedbacks associated with enhanced pCO_2 , although feedback magnitudes are considerably weaker than over land.

CO_2 -induced temperature change affects the atmospheric carbon budget and, although globally positive,

exhibits local regions of both signs and considerable geographical structure, especially over land. Increasing temperatures that favor plant growth also favor respiration so the sign and magnitude of the feedback is not uniform. Positive carbon–temperature feedbacks are found over most tropical land, while mid–high-latitude land exhibits negative feedbacks for this reason. There are also regions of positive and negative oceanic carbon–temperature feedback, in particular in the eastern tropical Pacific.

The geographical patterns of carbon–concentration and carbon–temperature feedbacks are reasonably robust across the three different simulations analyzed. The magnitudes of the feedbacks are somewhat less robust and scale nonlinearly as a consequence of the large CO₂ concentration changes engendered by the emission scenarios. The feedback patterns nevertheless serve to illustrate and quantify the carbon feedback processes operating in the modeled climate system.

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